

THE IMPACT OF TEACHING AIDS ON STUDENTS LEARNING IN SCIENCE

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the requirements for the degree of
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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

**Govt. College of Education F.B. Area,
Karachi**
DECLARATION

I, **HUMA D/O LIAQUAT ALI**, student of B.Ed. (Hons.) Elementary, hereby declare that no part of the action research report titled **“THE IMPACT OF TEACHING AIDS ON STUDENTS LEARNING IN SCIENCE ”** presented by me has been plagiarized from anywhere. Proper references are cited wherever necessary. I understand that the University of Karachi reserves the right to cancel the degree if any of the above declarations is proved false before or even after award of degree.

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CERTIFICATE OF APPROVAL

HUMA D/O LIAQUATALI, student of B.Ed. (Hons.) Elementary, has completed the Action Research entitled “**THE IMPACT OF TEACHING AIDS ON STUDENTS LEARNING IN SCIENCE** ” as per the scholarly standards set in the research project course guide provided by University of Karachi.

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DEDICATION

I dedicate this work to my beloved parents, whose support and inspiration have been my greatest strength throughout the journey of developing the ideas and concepts in this research project. Their love and care have been the foundation of this endeavour, without which it would not have been possible.

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ABSTRACT

This action research investigated the impact of teaching aids on students' learning, motivation, and interest in science at the grade eight level. The study was conducted at one of the schools of F.B. Area, Karachi, with six randomly selected students. A quantitative approach within the action research framework was used across three instructional cycles. Each cycle introduced progressively enhanced teaching aids, ranging from charts and models to real-life materials and multimedia tools. Data were collected through pre-tests, post-tests, and observation checklists to measure improvement in students' performance and engagement. Results revealed a steady rise in mean scores from 10.3 in the pre-test to 18.6 in the final post-test, showing significant academic improvement. Observation data also indicated increased interest, participation, and motivation. The findings confirmed that the thoughtful integration of teaching aids makes learning more interactive, meaningful, and enjoyable. It is recommended that science teachers regularly use varied teaching aids and incorporate reflective practices to foster student-centered learning.

CHAPTER 01

INTRODUCTION

1.1. BACKGROUND OF THE STUDY:

Science education is fundamental for national development, equipping learners with knowledge and skills to solve problems, innovate, and contribute to technological advancement. In developed countries, science education has evolved rapidly with interactive and student-centered methods (Idin, S. 2020). However, in Pakistan, many schools particularly in rural areas still rely on teacher-centered approaches, where students passively copy notes from the chalkboard. This method often hinders deep understanding and limits critical thinking. Reports from 2020–2023 reveal that more than 40% of middle schools learners in Pakistan fail to achieve satisfactory grades in science subjects (Shah et al 2024). A key factor is the shortage of effective teaching aids, which prevents students from fully grasping complex concepts. When learners are only exposed to lectures and textbooks, science appears abstract and difficult. On the other hand, teaching aids such as diagrams, experiments, animations, and models make science tangible and engaging and can visually represent information. Teaching aids play a massive role in teaching young students. At the elementary level, many topics can be difficult to figure out just from text. That's why models, diagrams, and videos are very useful. They make complex concepts easier for the betterment of students' academic performance, and learning becomes much clearer and more exciting by them. When students view a colorful picture of a flower or observe a video related to the topic, it catches their attention and helps them remember the information. According to Qasserras, L. (2024) Teaching aids also provide help to various types of learners, particularly those who are visual learners, as opposed to reading or hearing.

1.2. PURPOSE OF THE STUDY:

The main purpose of this study was to examine the impact of using teaching aids on the learning of grade 8th students in science. This research aimed to find out how different types of teaching aids, such as charts, models, experiments, and real-life materials, contributed to students' understanding and engagement. The study sought to identify whether the use of these aids improved students' comprehension of scientific concepts, enhanced their interest in the subject, and led to better performance in assessments. The findings helped teachers, school administrators, and curriculum planners understand the importance of incorporating teaching aids effectively to make science learning more interactive, meaningful, and result-oriented for middle school students.

1.3. STATEMENT OF THE PROBLEM

In many public schools across Pakistan, especially in rural and under-resourced areas, students struggled to understand and perform well in science subjects. One of the major reasons for this performance was the limited or ineffective use of teaching aids in classrooms. Science was a subject that required observation, experimentation, and visualization of concepts. However, when lessons were delivered only through verbal explanations or textbook reading, students found it difficult to grasp abstract ideas and scientific processes. The absence of visual aids, models, charts, and hands-on materials reduced students' interest, motivation, and understanding. As a result, many students failed to connect theory with practice, which ultimately affected their retention, curiosity, and academic achievement in science.

1.4. OBJECTIVES:

Specifically, the study aimed:

1. To examine the influence of teaching aids on students' interest and motivation in learning science.
2. To compare the performance of students taught with and without teaching aids.

1.5. RESEARCH QUESTIONS:

1. What is the influence of teaching aids on students' interest and motivation in learning science at the grade eight level?
2. What is the difference in the performance of grade eight students taught with and without the use of teaching aids?

1.6. SIGNIFICANCE OF THE STUDY

This study was significant because it emphasized the important role of teaching aids in enhancing students' learning experiences in science, particularly at the grade eight level. By examining the impact of various teaching aids, such as charts, models, experiments, and multimedia resources, the research helped to understand how these tools could make abstract scientific concepts easier to grasp. The study highlighted how the use of visual and practical materials not only improved students' comprehension but also increased their motivation, curiosity, and active participation during science lessons.

Moreover, the findings of this study provided valuable guidance for teachers by encouraging them to adopt more interactive and student-centered teaching methods. It also helped school administrators and policymakers in Pakistan recognize the need to allocate sufficient resources for teaching materials and training programs that promoted effective classroom practices.

Additionally, the study offered insights for curriculum developers to design science lessons that included hands-on and visual components, making learning more meaningful and reducing students' dependence on rote memorization. Ultimately, this research aimed to contribute to improving the overall quality of science education and fostering a deeper understanding of scientific principles among middle school students.

1.8. DEFINITIONS OF KEY TERMS:

Science Education:

Science education is the practice of teaching and learning science, which has been influenced by thinkers like (Thomas Kuhn).

Teaching Aids:

According to S.K. Kochchar, Any device that helps an instructor transmit facts, skills, attitudes, knowledge, understanding, and appreciation to a learner.

Academic Performance:

Narad and Abdullah (2016) define it as the knowledge acquired by a student that is assessed by a teacher through grades or by evaluating if educational objectives have been met.

Learning Outcomes:

Learning Outcomes are specific statements of what students should know and be able to do as a result of learning (Morss and Murray, 2005).

Chapter 02

Literature Review

Science education is regarded as one of the pillars of national development, as it equips learners with the knowledge and skills necessary to drive innovation, economic growth, and social progress. In Pakistan, however, science education continues to face serious challenges such as outdated teaching methods, lack of resources, and minimal use of teaching aids. These challenges result in poor comprehension, lack of interest, and low achievement among secondary school students (Shah, Iqbal, & Rauf, 2010). This chapter reviews previous studies and theoretical perspectives on the role of teaching aids in improving science learning outcomes.

Several studies have highlighted that the performance of Pakistani students in science subjects remains below international standards. Reports by the Board of Intermediate and Secondary Education (BISE) show that less than 50% of students achieve satisfactory grades in science subjects at the middle and secondary levels (Soomro, Qaisrani, & Uqaili, 2011). Many of these students struggle because teaching practices remain largely teacher-centered and heavily dependent on textbooks and rote memorization (UNESCO, 2012). In most public schools, teachers lack access to laboratories, projectors, or modern teaching aids, which limits their ability to demonstrate scientific concepts effectively. For example, explaining processes such as photosynthesis or the circulation of blood without visual or experimental support makes the content abstract and couldn't relate to learners. This often results in disengagement and poor performance (Shah & Khan, 2015). Given these challenges, it becomes essential to explore

alternative instructional strategies that can make science learning more meaningful and effective, one of which is the use of teaching aids.

Teaching aids are instructional materials and tools that support teachers in presenting lessons in an engaging, interactive, and practical manner. These include models, charts, laboratory equipment, videos, animations, and simulations. Research has consistently shown that teaching aids make science more understandable and memorable for students. According to Singh, Sharma, and Upadhyaya (2008), teaching aids reduce the workload of teachers by making explanations clearer while simultaneously enhancing student learning. They activate multiple senses, sight, hearing, and touch, resulting in improved concentration and retention. Similarly, Jadal (2011) emphasizes that audiovisual aids capture students' attention and sustain their interest throughout the lesson, making learning more enjoyable and meaningful. In the Pakistani context, Shah et al. (2010) found that science students taught with the aid of models, experiments, and visual tools performed significantly better in examinations compared to those taught through lectures alone. This clearly indicates that teaching aids are not merely supplementary but play a central role in strengthening students' grasp of scientific ideas. This suggests that teaching aids directly influence academic performance by transforming abstract scientific concepts into concrete, observable experiences.

Motivation and interest are crucial factors in science education. Ibe (2016) defines interest as the emotional engagement a learner feels towards a subject, which strongly correlates with achievement. Studies show that when science is taught using visual and interactive methods, students develop curiosity and enthusiasm, leading to better performance (Aniweze, 2014). In Pakistan, where many students perceive science as a difficult subject, teaching aids play a vital role in making learning enjoyable. For example, the use of a human skeleton model in biology

CHAPTER 02

LITERATURE REVIEW

classes or a simple electric circuit kit in physics sparks curiosity and encourages students to ask questions.

Kareem (2018) demonstrated that multimedia-based science instruction improved students' motivation levels and academic results, proving more effective than traditional chalk-and-talk methods. Thus, teaching aids not only enhance understanding but also reshape students' attitudes toward science, making the subject more approachable and stimulating.

Science education aims not only at knowledge acquisition but also at nurturing higher-order thinking skills such as analysis, evaluation, and problem-solving. Teaching aids, particularly experiments and project-based learning activities, provide students with opportunities to think critically. Prasad (2005) argues that when learners interact with visual and hands-on materials, they are more likely to develop problem-solving abilities and creativity. Similarly, Mangal (2008) highlights that instructional tools encourage inquiry-based learning, where students explore, experiment, and draw conclusions on their own. In Pakistan, this approach is especially important to move away from rote learning and towards fostering innovation and practical application of scientific knowledge. Therefore, the effective integration of teaching aids goes beyond improving academic results; it prepares students for lifelong learning and innovation.

Group activities involving teaching aids also enhance peer-to-peer learning and teamwork. Jimoh (2009) observed that when students worked together on science projects using audiovisual tools, they developed stronger social and communication skills. In Pakistani classrooms, collaborative use of teaching aids such as conducting experiments in groups helps build cooperation, leadership, and confidence among learners. This social dimension of learning further strengthens the role of teaching aids by creating a supportive environment where students learn not only from teachers but also from one another.

Globally, the effectiveness of teaching aids in science education has been widely documented. Studies in Nigeria (Ibe & Abamuche, 2019) and Turkey (Usak et al., 2009) revealed that students exposed to multimedia and visual aids performed significantly better in biology compared to those taught with lecture-only methods. In developed countries, multimedia simulations and interactive software are used to simplify complex scientific processes such as genetics, chemical bonding, and astronomical phenomena (Aloraini, 2012). These global examples provide valuable lessons for Pakistan, showing that the integration of modern teaching aids is not a luxury but a necessity for improving science education outcomes. These findings reinforce the idea that teaching aids are essential for effective science education across different contexts.

CHAPTER 03

METHODOLOGY

CHAPTER 03

METHODOLOGY

3.1. RESEARCH DESIGN:

In the action research framework this study followed a quantitative method. The method helped in gaining a deep understanding of students' thoughts, feelings, and learning experiences. As an action research project, it enabled the researcher to identify a real classroom problem, implement strategies to address it, and reflect on the results to improve teaching and learning.

According to Kemmis and McTaggart (2005), action research had evolved through various traditions and was recognized as a well-established and respected methodology in educational research. It promoted a culture of continuous learning and professional growth, as educators actively participated in the research process to enhance their instructional methods and classroom practices. This approach was particularly valuable for teachers who aimed to improve their teaching in a reflective, practical, and meaningful way.

Action research enables educators to critically analyze and reflect on their methods by combining systematic inquiry with real-world application. This resulted in data-driven decisions that increased the effectiveness of instruction and improved student learning outcomes.

3.2. RESEARCH SITE AND PARTICIPANTS:

The site for this action research project was one of the schools in F.B. Area, Block 15, Karachi. Six randomly selected students of 8th grade participated in the study.

3.3 RESEARCH INSTRUMENT:

I used the following research instruments for this action research project:

- Pre-test and Post-test
- Observation Checklist

3.4. PROCEDURE OF DATA COLLECTION (ACTION PLAN):

The present research project followed action research method and the data collection was cyclic. The data obtained from this study was helpful in understanding the impact of teaching aids on students' learning in Science.

3.5. RECONNAISSANCE: (THE PRELIMINARY STAGE):

Reconnaissance is the first step in identifying our current situation, our goals, and our anticipated path to success while conducting action research. In this preliminary study, the researcher collected data via pre-test to perform baseline assessment of students' performance and understanding of grade 8th students.

3.6. ACTION PLAN FOR CYCLE 01:

3.6.1 Teaching with basic teaching aids:

In the first cycle, the teacher introduced basic teaching aids like charts and diagrams to teach Science lessons. The main aim was to make lessons more understandable and engaging for students compared to traditional teaching.

3.6.2. Observation Checklist:

In Cycle 1, the observation checklist was used to assess students' performance in four key areas:

- (1) Interest and curiosity
- (2) Attention and focus
- (3) Participation and collaboration
- (4) Motivation and confidence.

During this cycle, basic teaching aids such as charts and diagrams were used to make science lessons more engaging. The checklist was completed by a critical friend to ensure fairness and reliability. Observations indicated a moderate level of improvement in students' interest and participation, showing that even simple teaching aids helped capture students' attention and encouraged classroom involvement.

3.6.3. Post-test:

After completing the lessons, a post-test was conducted to assess any progress in learning. The results revealed minimal improvement, showing that students found it difficult to understand Science concepts with basic resources. This confirmed the need for more interactive and visual teaching strategies in subsequent cycles.

3.7. ACTION PLAN FOR CYCLE 02:**3.7.1 Integration of enhanced Teaching Aids:**

In the second cycle, the teacher introduced a variety of teaching aids such as models and diagrams and real-life objects to make lessons more interactive and meaningful. These aids were selected carefully to match the content of the science topics being taught.

3.7.2 Observation Checklist:

In Cycle 2, the same observation checklist was used to record students' progress in all four areas: interest and curiosity, attention and focus, participation and collaboration, and motivation and confidence. This cycle involved the use of enhanced teaching aids, including models, diagrams, and real-life objects. The checklist was filled in by a critical friend to ensure objectivity and accuracy. The observations showed noticeable improvement in students' motivation, attention, and participation compared with Cycle 1, indicating that enhanced teaching aids made learning more engaging and effective.

3.7.3 Post-Test:

A post-test was again conducted to evaluate improvement in students' learning. Results showed noticeable progress in understanding, attention, and enthusiasm compared to Cycle One.

3.8. ACTION PLAN FOR CYCLE 03:**3.8.1 Enhanced Use of Teaching Aids:**

Based on the success of Cycle Two, the third cycle focused on improving the quality and consistency of teaching aid use. The teacher planned more student-centered lessons, involving multimedia presentations and group-based activities where students interacted and explored scientific ideas.

3.8.2 Observation Checklist:

In Cycle 3, the same observation checklist was again used to assess students' progress in the areas of interest and curiosity, attention and focus, participation and collaboration, and motivation and confidence. This cycle involved the use of multimedia presentations and group-based activities to make science learning more practical and interactive. The checklist was completed by a critical friend to ensure accurate and unbiased observations. Results

showed a high level of improvement in all areas, as students displayed greater enthusiasm, active participation, and confidence while engaging in practical activities.

3.8.3. Post-Test:

A final post-test was conducted to measure overall improvement in learning. The results showed a significant increase in comprehension, motivation, and participation, confirming that the consistent and purposeful use of teaching aids had a strong positive impact on students' learning in science.

3.9. PROCEDURE OF DATA ANALYSIS:

3.9.1 Analysis of Post-Test, and Observation Checklist:

In all the three cycles the data was collected from a post-test tool subjected to a quantitative analysis. A checklist also used as a complementary tool with all post-test. The responses' mean was used to determine the data's central tendency. A numerical representation of students' learning was given by the mean. The teacher researcher was able to assess the impact of teaching aids on students learning in Science by calculating the average response for all post-tests.

3.10 ETHICAL CONSIDERATION:

Ethical considerations were carefully observed throughout the entire research process to ensure that the rights, dignity, and well-being of all participants were fully protected. Before the commencement of the study, formal permission was obtained from the school administration to conduct the research within the school premises. The researcher clearly explained the purpose, procedures, and expected outcomes of the study to all participants in simple and understandable language appropriate for their age level.

- Participation in the study was completely voluntary. Students were informed that they had the right to withdraw from the study at any stage without facing any pressure, penalty, or negative consequences. No student was forced or influenced to take part in the research activities.
- To protect participants' privacy, confidentiality and anonymity were strictly maintained. Students' names and any personal information were not recorded or disclosed in any part of the study or its results. All collected data were used only for academic purposes and kept secure by the researcher.
- The researcher made sure that the study environment remained respectful, non-judgmental, and emotionally safe for all participants. The teaching and observation processes were conducted with sensitivity to students' comfort levels and learning needs. Additionally, all materials and activities used during the research were age-appropriate and designed to support learning rather than cause stress or discomfort.

Chapter 04

DATA ANALYSIS

CHAPTER 4

DATA ANALYSIS

This chapter contains the results of the study including mean comparison and reflections over observation checklist, in three cycles of action research. Quantitative data were collected for this study from 6 students of grade 8 at one of the school of F.B. Area, karachi.

4.1 DEMOGRAPHIC INFORMATION:

4.1.1 TABLE: DEMOGRAPHIC

NO OF PARTICIPANTS	TEACHING IN GRADE/CLASSES	Gender		AGE MEAN
		Male ○	Female ●	
6	8th			13.2

4.2 DATA ANALYSIS (CYCLIC):

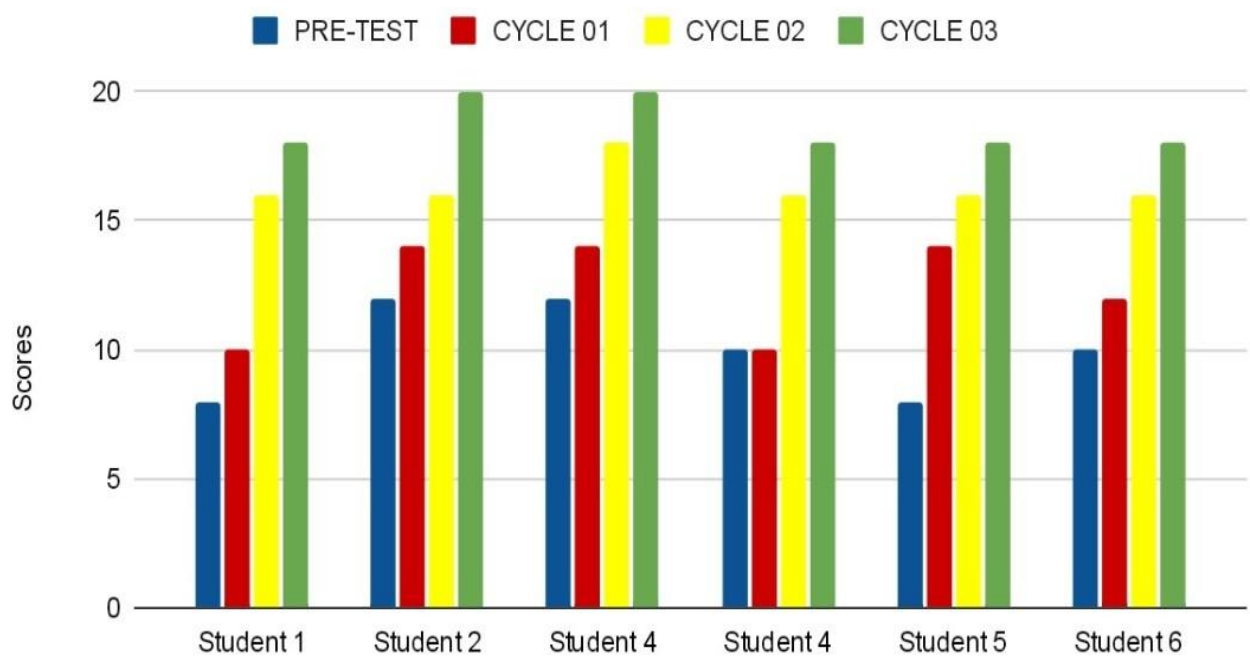
4.2.1 TABLE: MEAN SCORES (CYCLE WISE)

S.No	STUDENTS NAME	PRE-TEST	CYCLE 01	CYCLE 02	CYCLE 03
1	Student 1	08	10	16	18
2	Student 2	12	14	16	20
3	Student 3	12	14	18	20
4	Student 4	10	10	16	18
5	Student 5	10	14	16	18
6	Student 6	10	12	16	18
i	Total	62	72	98	112

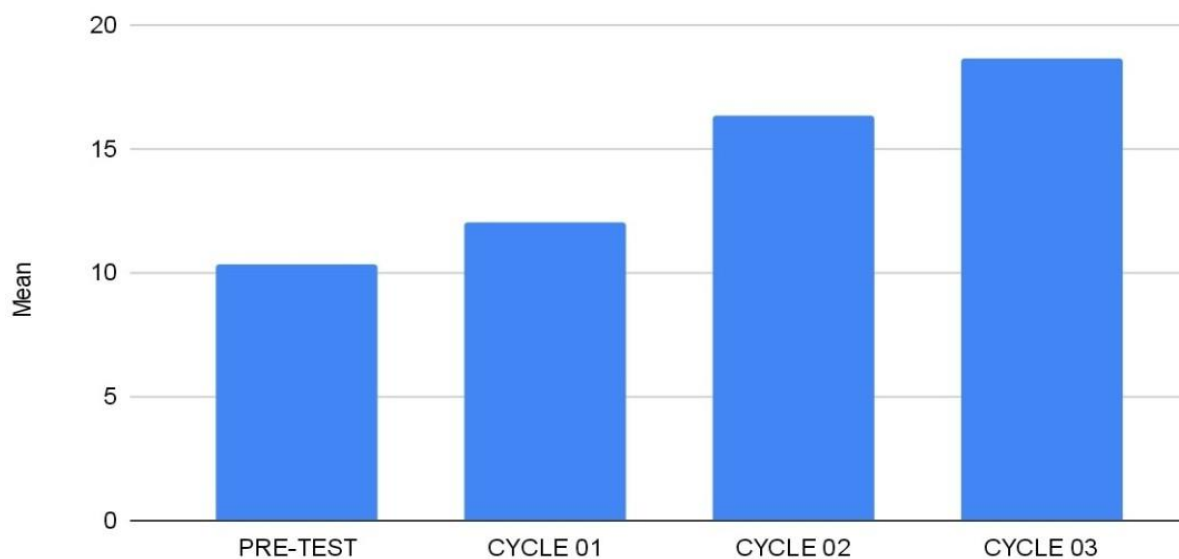
ii	Mean	10.3	12	16.3	18.6
iii	Percentage	51.6%	60%	86.6%	93.3%

I used a test to assess improvement in students' learning in science as a result of using teaching aids. The numeric values are given in the form of a graph given below:

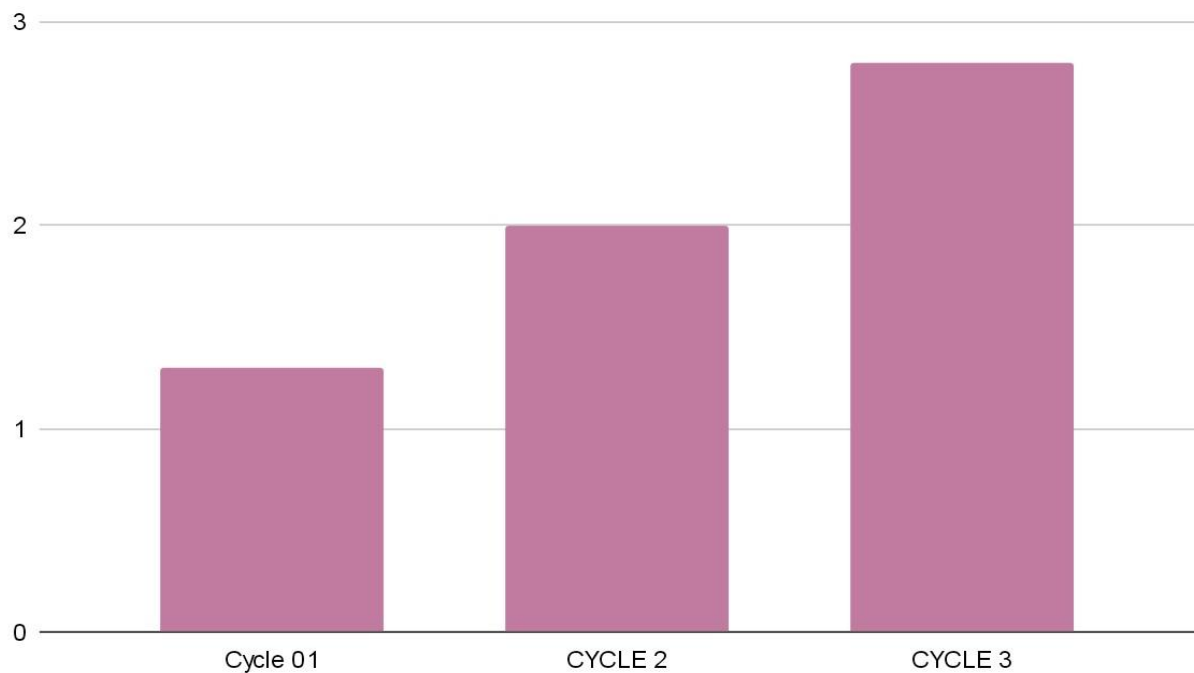
4.2.2 FIGURE: INDIVIDUAL MEAN SCORE



4.2.3 FIGURE: GROUP MEAN SCORE



4.2.4 PROGRESS MEAN OF OBSERVATION CHECKLIST



4.3 THE PRELIMINARY STAGE (PRE-TEST):

Before starting Cycle One, I conducted a pre-test to assess the level of existing knowledge and understanding of grade 8th students. The average mean obtained from this pre-test was 10.3. This initial evaluation helped establish a baseline understanding of the students' learning and understanding in science.

4.4 OBSERVATION CHECKLIST:

In this study, I used an observation checklist as my second research tool to closely observe and understand how my students responded to the use of teaching aids during science lessons. The main purpose of this tool was to examine how different teaching aids influenced students' interest, motivation, and participation during classroom activities.

The observation checklist was filled by my critical friend, who observed each lesson carefully and recorded the students' responses and behaviors. This approach helped ensure that the observations were objective and unbiased, as the checklist was completed by someone other than me. My critical friend and I later discussed the observations together to gain a clearer and more balanced picture of students' engagement during each lesson.

The checklist consisted of 10 key items, each designed to assess a particular aspect of students' motivation and classroom behavior. It focused on how students interacted with teaching aids, how attentive and curious they were, and how willing they were to participate in class activities.

A 3-point Likert scale was used for each item, where:

3 = Always, 2 = Sometimes and 1 = Rarely.

This simple scale made it easy to assign numerical values to the behaviors observed, allowing me to later calculate average scores for analysis. The items included the following statements:

1. Students show curiosity and excitement when teaching aids are used in the science lesson.
2. Students pay close attention during lessons supported by teaching aids.
3. Students participate actively in class discussions and activities related to teaching aids.
4. Students ask relevant and thoughtful questions during the lesson.
5. Students express enjoyment while observing or handling teaching aids.
6. Students show motivation to complete tasks or experiments using teaching aids.
7. Students demonstrate focus and attentiveness throughout the lesson.
8. Students cooperate with peers during science activities involving teaching aids.
9. Students show confidence while explaining or presenting their understanding of concepts.
10. Students display positive attitudes toward learning science through teaching aids.

In each research cycle, two observations were conducted, one during the early part of the lesson and another toward the end, using the same checklist. The two scores were then averaged to obtain the mean score for the entire cycle. This allowed me to identify any changes in student behavior and engagement as the lesson progressed. The results were interpreted using the following guide:

1.00 – 1.49 = Low

1.50 – 2.49 = Moderate

2.50 – 3.00 = High

When comparing results across all three cycles, I found that the mean scores consistently improved, showing a gradual but steady rise in students' motivation and participation. These results clearly indicated that the use of teaching aids had a positive and meaningful impact on students' classroom involvement. With each new cycle, as I introduced more interactive and visually appealing materials, students became more curious, confident, and eager to learn. It was rewarding to see how the right use of teaching aids could transform both the learning environment and students' attitudes toward science.

4.5 CYCLE 01:

I completed my initial cycle in two days, working with 6 students in 8th Grade. In the first cycle of my research, my main aim was to see how the use of teaching aids could affect my students' motivation and interest in learning science. During this phase, I used simple materials such as charts and diagrams to support my lesson. At the start, these visuals caught the students' attention, and most of them seemed curious and ready to participate. They answered questions eagerly and showed excitement about the topic.

However, as the lesson went on, I noticed that their interest began to fade. Some students lost focus, started chatting with their classmates, and seemed less involved. Although a few remained attentive, the overall energy in the classroom dropped as the lesson progressed. This made me realize that while teaching aids helped capture their attention initially, they weren't strong enough to keep it throughout the session.

After teaching, I conducted a post-test to check how much they had understood and learned from the lesson. I also used an observation checklist to note their participation, attentiveness, and overall motivation.

4.5.1 Results of the Post-test:

In the first cycle, the mean score of the post-test was 12, which was 60% of the total marks. This showed that students understood some concepts but their overall learning was limited. Although the use of simple teaching aids like charts and diagrams caught their attention at first, many students lost focus as the lesson continued. Their interest and motivation were inconsistent, and this was reflected in their moderate post-test performance.

4.5.2 Observation checklist:

In the first cycle, the mean score of the observation checklist was 1.3, which falls in the low range of engagement. This showed that most students were not very attentive or motivated during the lesson. Although basic teaching aids like simple charts and diagrams were used, they were not enough to keep students interested. Many lost focus and only participated when prompted. Overall, their engagement and enthusiasm were quite limited during this cycle.

4.6 CYCLE 02:

After reflecting on the outcomes of Cycle 1, I realized that my teaching methods needed to be more engaging and visually appealing. So, in the second cycle, I improved my materials by using models, diagrams and real-life objects that were directly related to the key points of the lesson. My goal was to make the lesson livelier and easier for students to connect with.

This time, I encouraged my students to be more involved. I asked open-ended questions, invited them to share their ideas, and guided them to interpret what they saw on the charts. The classroom atmosphere became much more interactive and enjoyable. I noticed that students stayed focused for longer periods, and there was a significant drop in off-task behavior. They looked more confident and eager to participate, and the discussion became more balanced, with more students contributing their thoughts.

At the end of the lesson, I again gave a post-test to measure their understanding and progress. I reused the same observation checklist to record their engagement and motivation, and my critical friend filled in the observation checklist again.

4.6.1 Results of the Post-test:

In the second cycle, I improved my teaching aids by using models, diagrams and real-life objects to make the lesson more engaging. As a result, students became more active, attentive, and motivated during the class. The mean score increased to 16.3 (86.6%), showing clear improvement in understanding and participation. The enhanced visuals helped students grasp key ideas more effectively and stay involved throughout the lesson.

4.6.2 Observation checklist:

In the second cycle, the mean score improved to 2.0, which indicates a moderate level of engagement. The introduction of teaching aids, such as models, diagrams and real-life objects made the lesson more interesting and interactive. Students showed better attention, participated more actively, and demonstrated growing motivation to learn. This improvement reflected that the enhanced visuals had a positive effect on their classroom behavior and learning interest.

4.7 CYCLE 03:

By the third cycle, I wanted to make my lesson even more engaging and interactive. I decided to use multimedia presentations and group-based activities that highlighted the main ideas of the topic. I also involved students during the lesson by questioning technique while explaining the main idea. I allowed students to share their concepts and thoughts in their own words. And then I divided students into groups and assigned some group-based activities to enhance their understanding and to grasp knowledge on their own.

Throughout this lesson, I noticed a remarkable change in students' attitudes. They were alert, curious, and genuinely excited to learn. Many of them asked thoughtful questions and even shared their own observations. The use of multimedia presentations especially grabbed their attention. I could see the excitement on their faces when they connected what they saw to what they had previously learned. The whole environment felt energetic and positive, with every student participating in one way or another.

After the lesson, I conducted another post-test to assess their understanding and learning outcomes, and my critical friend filled in the observation checklist once again.

4.7.1 Results of the Post-test:

In the third cycle, I introduced interactive aids such as multimedia presentations, group-based activities. Students showed great excitement and curiosity, asking thoughtful questions and staying focused throughout the lesson. Their post-test results showed the highest progress, with a mean score of 18.66 (93.3%). This confirmed that using interactive and hands-on teaching aids had a strong positive effect on students' motivation and learning in science.

4.7.2 Observation Checklist:

In the third cycle, the mean score further increased to 2.8, which falls within the high range of engagement. This showed a significant improvement in students' motivation, participation, and overall classroom involvement. The use of multimedia presentations and group-based activity materials made the lesson more engaging and enjoyable. Students were curious, confident, and excited to take part in class discussions and activities. The high mean score clearly indicated that teaching aids had a strong positive impact on students' learning attitudes and enthusiasm for science.

4.8 REFLECTIONS:

I conducted my action research with 6 students of Grade 8 to examine how the use of different types of teaching aids could influence their learning, motivation, and interest in Science. The research was completed in three cycles, each spanning two days. During each cycle, I planned, implemented, observed, and reflected on the effectiveness of various teaching aids, while continuously improving my approach based on the outcomes of the previous cycle.

In the first cycle, I used basic teaching aids such as simple charts and diagrams to teach the science topic. At the beginning of the lesson, students seemed curious, but their attention and participation gradually decreased as the class continued. Some students were easily distracted, and only a few responded during discussions. I noticed that while the use of basic visuals created some level of understanding, it failed to sustain students' motivation. The results of the post-test and observation checklist confirmed this observation, showing low interest and limited engagement among students. This cycle helped me realize the need for more appealing and interactive teaching materials to maintain students' focus and enthusiasm throughout the lesson.

In the second cycle, I revised my teaching strategy by using enhanced teaching aids such as models, diagrams and real-life objects. These visuals made the lesson more engaging and easier for students to understand. I also involved students more actively by asking questions and encouraging them to relate the visuals to real-life examples. This time, the classroom atmosphere was more lively and interactive. Students showed improved attention, better motivation, and greater willingness to participate in class activities. The post-test and observation checklist scores improved compared to the first cycle, indicating that enhanced teaching aids had a positive impact on students' learning and interest in Science.

In the third cycle, I further improved my teaching approach by using multimedia presentations and group-based activities to provide students with a more hands-on and interactive learning experience. I allowed students to share their understanding and concepts in their own words. This approach transformed the classroom into an active learning space where students were excited, curious, and eager to explore the lesson. Their engagement remained high throughout the session, and they asked thoughtful questions that showed a deeper understanding of scientific concepts. The results of the post-test and observation checklist in this cycle showed a significant increase in performance, interest, and motivation compared to the first and second cycles.

Through these three cycles, I realized that the effective use of teaching aids plays a vital role in improving students' learning outcomes in Science. As I refined my teaching strategies and introduced more interactive materials, students' engagement and understanding grew remarkably. The action research experience not only improved my teaching practice but also demonstrated how meaningful and enjoyable learning can become when students are actively involved through creative and well-planned teaching aids.

CHAPTER 05

CONCLUSION, DISCUSSION AND RECOMMENDATIONS

CHAPTER 5: DISCUSSION, CONCLUSION AND RECOMMENDATIONS

This present chapter discusses the findings of the study related to previous studies. It shows the contribution of the present research to the literature as well as academic practices, along with the conclusion, implications of the study, and ends with the limitations of the study.

5.1 DISCUSSION:

This action research was carried out to explore how the use of different types of teaching aids including charts, models, real-life objects, and multimedia affects students' learning, motivation, and interest in science at the grade eight level. The study was implemented in three cycles, each designed to progressively enhance the variety and interactivity of teaching aids used. Through continuous reflection, observation, and testing, the researcher observed how students' attitudes, engagement, and academic performance changed as the instructional strategies evolved.

At the beginning of the research, during the pre-test phase, students' performance and participation were relatively low. This reflected their limited exposure to interactive learning experiences and their dependence on traditional lecture-based teaching. The mean score of 10.3 in the pre-test suggested that students had only a basic understanding of scientific concepts and were not deeply engaged in learning.

In Cycle 1, the introduction of basic teaching aids such as charts and diagrams generated initial curiosity but failed to maintain long-term engagement. Students were interested at first, yet their focus declined as the lesson progressed. The post-test results showed only slight improvement, with the mean rising to 12 (60%). The observation checklist revealed a low

engagement level (mean score 1.3), confirming that while visuals helped attract attention momentarily, they were insufficient to sustain motivation or enhance deep understanding. This highlighted the limitation of using simple visual aids without interactive elements.

During Cycle 2, the researcher expanded the variety of teaching aids by including models, diagrams, and real-life objects. This shift had a noticeable positive effect. Students became more involved and attentive, actively participated in discussions, and demonstrated greater enthusiasm. Their mean post-test score rose to 16.3 (86.6%), and the observation checklist indicated a moderate engagement level (mean score 2.0). These results aligned with previous studies (e.g., Kareem, 2018; Singh et al., 2008) showing that hands-on and visual learning materials enhance comprehension and motivation. The students' growing confidence and curiosity during this phase reflected their improved conceptual understanding and enjoyment of science.

The most significant progress was observed in Cycle 3, where multimedia presentations and group-based activities were integrated into the lessons. The classroom atmosphere became highly interactive, students asked meaningful questions, discussed ideas freely, and worked collaboratively. Their learning became active rather than passive. The post-test mean score increased to 18.6 (93.3%), and the observation checklist mean reached 2.8, showing a high level of engagement. These findings strongly suggested that interactive and student-centered teaching aids not only improve academic performance but also transform students' attitudes toward science.

The gradual improvement across all three cycles clearly demonstrated that the type and quality of teaching aids directly influenced students' learning outcomes. The findings confirmed that

visual and hands-on experiences make abstract scientific concepts concrete and easier to understand. When students see, touch, or interact with learning materials, they develop stronger connections between theory and practice. Moreover, the increased motivation and collaboration observed in later cycles showed that teaching aids contribute to creating a supportive, engaging, and enjoyable learning environment.

The results of this study are consistent with global research findings, such as those of Usak et al. (2009) and Ibe & Abamuche (2019), who found that students exposed to multimedia and visual learning tools performed significantly better than those taught through lectures alone. In this study, students not only improved academically but also developed confidence, curiosity, and teamwork skills, qualities essential for effective learning in science.

Overall, the discussion highlights that teaching aids are not just supplementary tools but powerful pedagogical resources that can bridge the gap between abstract content and real-world understanding. Their thoughtful and consistent use can transform traditional classrooms into vibrant learning spaces where students actively construct knowledge and develop a love for learning.

5.2 CONCLUSION:

The findings of this action research strongly indicate that the use of teaching aids has a significant positive impact on students' learning, motivation, and academic performance in science. Over the three cycles, as the variety and interactivity of teaching aids increased, students showed remarkable progress in understanding, attention, and participation.

The research revealed that:

1. Teaching aids make learning more concrete, visual, and meaningful.
2. Students taught with interactive teaching aids perform better than those taught with traditional methods.
3. The use of teaching aids enhances students' motivation, curiosity, and confidence in learning science.
4. Active engagement through visual and hands-on materials leads to better comprehension and retention of scientific concepts.

This study also demonstrated that effective science teaching goes beyond delivering content, it involves creating experiences that connect knowledge with exploration and discovery. When students are given the opportunity to see, manipulate, and discuss materials, they learn not only facts but also how to think scientifically.

In conclusion, teaching aids play a vital role in transforming classroom learning into an engaging, meaningful, and student-centered experience. Their systematic and creative use should therefore be considered an essential part of every science lesson, especially at the middle school level.

5.3 RECOMMENDATIONS:

Based on the findings and reflections from this action research, the following recommendations are proposed:

1. Teachers should integrate a variety of teaching aids, such as models, experiments, multimedia, and real-life objects into daily science lessons to make learning more interactive and enjoyable.
2. Schools and education departments should organize workshops and training sessions to equip teachers with the skills needed to design, select, and use teaching aids effectively.
3. School administrators and policymakers should allocate adequate funds to ensure that classrooms are well-equipped with essential teaching materials like charts, models, projectors, and lab apparatus.
4. Teachers should involve students in preparing and using teaching aids, as this promotes creativity, responsibility, and a deeper understanding of the subject matter.
5. Incorporating multimedia presentations, educational videos, and interactive simulations can further enhance engagement and make complex topics easier to grasp.
6. Curriculum planners should include guidelines that emphasize the use of visual and practical teaching strategies in science education to move away from rote memorization toward conceptual learning.
7. Teachers should adopt an action research approach to continuously reflect on their teaching methods and make necessary improvements based on student feedback and performance.

5.4 LIMITATIONS OF THE STUDY:

While the results of this research were encouraging, several limitations should be acknowledged:

- The study involved only six students from one school, which limits the generalization of findings to larger populations.
- The research was conducted over a short period of time, so long-term effects of teaching aids on learning retention and performance could not be fully observed.
- Differences in teaching techniques and classroom management may have influenced students' responses to the teaching aids.
- The availability of materials and technological tools was restricted, which may have limited the full potential of the intervention.
- The observation checklist and post-tests provided valuable insights but did not capture all dimensions of students' creativity, problem-solving, and emotional engagement.

Despite these limitations, the findings provide a strong foundation for understanding how teaching aids influence learning in real classroom settings.

5.5 FUTURE RESEARCH

This study opens several avenues for future research:

1. Future studies should involve a larger number of students from different schools and regions to validate and generalize the findings.
2. Research should be conducted over an extended period to examine the long-term impact of teaching aids on students' achievement, retention, and attitude toward science.

3. Future research could compare the effectiveness of specific types of teaching aids, such as multimedia vs. physical models, to identify which are most beneficial for different topics or age groups.
4. Further research could investigate teachers' attitudes, challenges, and competencies related to using teaching aids effectively in diverse learning environments.
5. Since teaching aids have proven effective in science, future research can explore their impact in other subjects like mathematics, social studies, or languages.

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APPENDIX A

Consent letter

 **Government College of Education**
Block 15, F. B. Area, Karachi
DDO code KQ-2149

NO. GCE/ - / 2024-2025 Dated: 27/09/25

The H.M/Principal,
Ibrahim Ali Bhai g.s.s,
F.B area, Block#15.

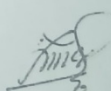
Subject: **PERMISSION TO CONDUCT ACTION RESEARCH AS PART OF**
B.Ed. (Hons.) 4 YEAR (ELEMENTARY) 8TH SEMESTER

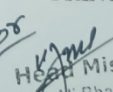
Dear Sir/Madam,

The action research is an extended activity of the teaching practice (Practicum) of the current B.Ed. curriculum. Mr./Ms. Huma Liaquat Ali is currently enrolled in this college as a student of B.Ed. (Hons.) 4 year (Elementary) 8th semester 2025. He/She is conducting a research entitled "The impact of Teaching aids on Students' Learning in Science." under the supervision of Prof. Tabassum Kausar.

Kindly allow him/her to implement innovative teaching strategies and/or to collect necessary data. During this process, he/she will strictly follow the research ethics. Your cooperation and facilitation in this academic activity will be highly appreciated.

Best Regards


PRINCIPAL

For

Head Mistress
Ibrahim Ali Bhai Elementary
Girls School No.4,
Gulberg Town, Karachi.

APPENDIX B

Observation Checklist

Topic: _____ Observer: _____ Grade: _____
 Duration: _____ Date: _____

INSTRUCTIONS: Observe the Performance of students during lecture and mark the column according to it.

No	Observation items	3	2	1
1	Students show curiosity and excitement when teaching aids are used in the Science lesson.			
2	Students pay close attention during lessons supported by teaching aids.			
3	Students participate actively in class discussions and activities related to teaching aids.			
4	Students ask relevant and thoughtful questions during the lesson.			
5	Students express enjoyment while observing or handling teaching aids.			
6	Students show motivation to complete tasks or experiments using teaching aids.			
7	Students demonstrate focus and attentiveness throughout the lesson.			
8	Students cooperate with peers during science activities involving teaching aids.			
9	Students show confidence while explaining or presenting their understanding of concepts.			
10	Students display positive attitudes toward learning Science through teaching aids.			

Total Score: ____

Scale: 3 = Always 2 = Sometimes 1 = Rarely

APPENDIX C

Action plans

(From Cycle 01)

LESSON PLAN NO _____

CLASS: VIII DATE: 04-Oct-2021

SUBJECT: Physics DURATION: 35 min

TOPIC: Types of Motion CLASS STRENGTH: 101

1. Specific Objectives:

By the end of the lesson students will be able to:

- Define motion in their own words.
- Describe the different types of motion. (Rotatory, Vibratory, Translatory)
- Identify & Classify examples correctly

2. Teaching Resources:

Chart (Showing three types of motion)

Images / Diagrams: (Fan, Pendulum, Car)

Evaluation sheet: MCQ's

3. Classroom Management Strategy:

Maintain engagement through Q/A technique

4. Teaching Method: Interactive Discussion Method

5. Procedure:

i) Motivation:

How does the fan moves when we switch it on? (Round)

Now, how does the pendulum of clock moves? (backward & forward)

Do both of these objects move in the same way or not (No they not)

What are these movements called in physics? (ignore)

ii) Announcement of Topic: Today we will learn about

"Motion and its Types"

Development:

Step # 01: Introduction of Motion

Teacher explains:

When an object changes its position with time it is said to be in "motion"

Step # 02: Explanation of Types of Motion

Now, Teacher will show diagrams & chart of types of motion and explain each type:

Rotatory Motion: "When an object moves around a fixed axis or center."

Examples: Fan blades, wheel of a car.

Vibratory Motion: "When an object moves back and forth rapidly around a fix point."

Example: Pendulum, vibrating ruler on desk.

Translatory Motion: "When an object moves from one place to another in a straight or curved path."

Example: A moving car, a ball rolling.

Board Summary/ Recapitulation:

Teacher reviews key points by asking quick oral questions:

- What is motion?
- Name of three types of motion we learned today?
- Give one example of each type.

Evaluation:

Teacher will distribute Test sheets for evaluation among students and ask them to answer each MCQs correctly.

Home Work:

List 5 objects from your surroundings and write which type of motion each one shows.

Reflections:

In, today's lesson students showed curiosity but struggled to stay focused throughout the lesson. I realized the need of more interactive aids and activities for the next plans to improve understanding.

(From Cycle 02)

LESSON PLAN NO 4

CLASS: VIII DATE: 06-Oct-2025

SUBJECT: Physics DURATION: 35 min

TOPIC: Matter and Its States CLASS STRENGTH: 10

1. Specific Objectives:

By the end of the lesson students will be able to :

- Define matter and its types
- Describe the characteristics of solids, liquids & gases.
- Identify examples of each state from daily life.

2. Teaching Resources:

- Model : Molecules in solid, liquid & Gas
- Diagrams : Chart showing arrangement of particles in different states
- Real-life objects : Ice-cube, water and air-filled balloon.

3. Classroom Management Strategy:

- Arrange students in a semicircle for better visibility of demonstration
- Move around the class to keep everyone engaged

4. Teaching Method: Demonstration & Discussion method.

5. Procedure:

i) **Motivation:** Teacher will place a pen and a book on desk and ask questions:

- (i) Which of these covers more space?
- (ii) Do both the things have same mass?
- (iii) Now, If i move both of them, will it make any difference?
- (iv) So what we call those things that occupies space & has mass?

ii) **Announcement of Topic:** "Today we will learn about matter & The States of matter."

iii) **Development:**

Step#01: Introduction (Definition of Matter)

Teacher Explains:

"Everything that has mass and occupies space is called matter."

Example: air, water, chair, books, etc.

Step#02:- Explanation of the three states of Matter

Teacher will demonstrate it by Ice cubes & water & air-filled balloons. and also use models to show molecular arrangement of these states.

Solids: Solids have a fixed shape and volume. Their particles are tightly packed.

Example: Ice cubes & Pencil

Liquid: Liquids have a fixed volume but no fixed shape. They take the shape of their containers.

Example: water, milk & Juice.

Gases: Gases have neither a fixed shape nor a fixed volume. Their particles are far apart and move freely.

Example: Air, Perfume vapors & fumes.

Step#03. Comparative Discussion.

Teacher will make table and add short points of each state and ask students to read & discuss example from daily life.

Board Summary/ Recapitulation:

Teacher asks quick oral questions:

- i- what are the three states of matter?
- ii- Which state of matter has a definite shape?
- iii- Why does the smell of perfume spread in a room?
- iv- Give one example of a solid, liquid & gas.

Evaluation:

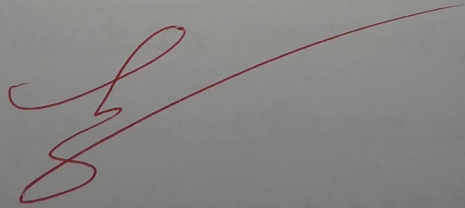
Teacher will distribute test sheets for evaluation among students and ask them to correctly answer each item of given test.

Home Work:

Observe your surroundings and list 3 examples each of Solids, liquids & gases. Draw their particle arrangement in your notebook.

Reflections:

In today's lesson students participated more actively and ~~showed~~ better understanding. Visual aids improved engagement. However, time management was a challenge, so I plan to give clear task limits in next class.



(From Cycle 03)

LESSON PLAN NO 6

CLASS: VIII DATE: 08-Oct-2025

SUBJECT: Chemistry DURATION: 35 min

TOPIC: Physical & Chemical changes CLASS STRENGTH: 10

1. Specific Objectives:
 By the end of lesson students will be able to .
 • Define physical & chemical changes.
 • Differentiate between Physical & chemical changes.
 • Identify examples of both types of changes in daily life.

2. Teaching Resources:
 • Multimedia presentation: To show picture of Examples
 • of physical & chemical changes.
 • Real-objects: Paper, candle, baking sod, vinegar.
 Charts for activity.

3. Classroom Management Strategy:
 • Arrange students in small groups (4-5 member)
 • Ensure all students can view the multimedia presentation
 • clearly.

4. Teaching Method: Multimedia Demonstration

5. Procedure:

i) **Motivation:**
 (i) What happens when you melt ice?
 (ii) What happens when you burn the paper?
 (iii) Can the both process be reversed? (iv) What we call
these changes in the language of Science?

ii) **Announcement of Topic:** Today we will study about
"Physical & chemical changes."

Step # 01: Introduction:

Teacher explain that matter can change in two main ways. Physical Or Chemical. Teacher use multimedia presentation / slide to show example of each type.

Step # 02: Teacher will Explain each type.

Physical Change: A change in shape, size or state without forming a new substance. Physical changes are usually reversible.

Example: Melting of ice, dissolving sugar in water.

Chemical change: A change that produces a new substance with different properties. Chemical changes are usually irreversible.

Example: Burning paper, rusting of iron, vinegar reacting with baking soda.

Activity:- • Teacher divide students into small-groups. Each group receive a cards showing scenarios (e.g: ice, melting, candle burning, rusting iron, etc)
• Students classify each as chemical or Physical change on Activity chart.

Board Summary/ Recapitulation:

- Q¹ - Teacher recap main points by quick questions.
Q² - What is the main different between P.C & C.C?
Q³ - Which type of change is reversible?
Q⁴ - Give two examples of each.

Evaluation:

Teacher will distribute the test sheets for evaluation among students and ask them to solve the test paper.

Home Work:

Observe 2 changes at home & write whether each is a physical or chemical change.

Reflections:

In today's class students displayed strong interest and confidence in answering questions. Learning outcomes improved and teamwork was evident. The use of teaching aids proved effective in enhancing both understanding & motivation.

APPENDIX D

POST-TEST SAMPLES

(From Cycle 01)

Post-Test:**Topic: Types of Motion****Name:** _____ **Date:** _____**Choose the correct answer for each question.**

Q1. When an object changes its position with time, it is said to be in:

- a) Rest. b) Shape. c) Motion. d) Force

Q2. The movement of a fan's blades shows:

- a) Translatory motion. b) Rotatory motion. c) Vibratory motion. d) None of these

Q3. A moving car on a straight road is an example of:

- a) Rotatory motion. b) Vibratory motion. c) Translatory motion. d) Circular motion

Q4. The motion of a tuning fork when struck is:

- a) Translatory. b) Rotatory. c) Vibratory. d) Linear

Q5. The Earth spinning on its axis shows:

- a) Rotatory motion. b) Translatory motion. c) Vibratory motion. d) Periodic motion

Q6. The motion that repeats to and fro around a fixed point is called:

- a) Circular motion. b) Vibratory motion. c) Rotatory motion. d) Translatory motion

Q7. When all parts of an object move in the same direction, it is:

- a) Translatory motion. b) Rotatory motion. c) Vibratory motion. d) Random motion

Q8. Which of the following shows both rotatory and translatory motion?

- a) Fan. b) Car wheel. c) Pendulum d) Ruler

Q9. Which of the following is not an example of rotatory motion?

- a) Ceiling fan. b) Clock hands. c) Swing. d) Earth on its axis

Q10. The movement of a ruler fixed at one end and plucked at the other end is:

- a) Translatory. b) Rotatory. c) Vibratory. d) Irregular

(From Cycle 02)**Post-Test:****Topic: Matter and Its States****Name:** _____ **Date:** _____**Choose the correct answer for each question.**

Q1. Anything that has mass and occupies space is called:

- a) Energy. b) Matter. c) Force. d) Atom

Q2. Which of the following has a definite shape and volume?

- a) Solid. b) Liquid. c) Gas. d) Plasma

Q3. Liquids have:

- a) Fixed shape and fixed volume. b) Fixed shape but no fixed volume
c) Fixed volume but no fixed shape. d) No fixed shape and no fixed volume

Q4. Which state of matter has particles that are very close together?

- a) Solid. b) Liquid. c) Gas. d) Plasma

Q5. In which state of matter do particles move freely and spread quickly?

- a) Solid. b) Liquid. c) Gas. d) None of these

Q6. The smell of perfume spreading across a room shows that:

- a) Liquids flow. b) Gases expand to fill space. c) Solids have fixed shape. d) Matter has weight

Q7. Which of the following changes is an example of a solid changing into a liquid?

- a) Freezing of water. b) Melting of ice. c) Evaporation of water. d) Condensation of steam

Q8. Which state of matter takes the shape of its container but keeps the same volume?

- a) Solid. b) Liquid. c) Gas. d) Plasma

Q9. The particles in solids are:

- a) Far apart and move freely. b) Tightly packed and vibrate in place
c) Loosely packed and slide past each other. d) Not arranged at all

Q10. Which of the following shows all three states of matter?

- a) Water, Ice, Steam. b) Milk, Juice, Oil
c) Rock, Sand, Clay. d) Air, Oxygen, Nitrogen

(From Cycle 03)**Post-Test:****Topic: Physical and Chemical change****Name:** _____ **Date:** _____**Choose the correct answer for each question.**

Q1. A change in which no new substance is formed is called:

- a) Chemical change. b) Physical change. c) Biological change. d) Natural change

Q2. Which of the following is a chemical change?

- a) Melting of ice. b) Boiling of water. c) Burning of paper. d) Breaking of glass

Q3. Which of the following is a physical change?

- a) Rusting of iron. b) Burning of candle. c) Tearing of paper. d) Cooking of food

Q4. The rusting of iron is a:

- a) Temporary change. b) Physical change. c) Reversible change. d) Chemical change

Q5. Which of the following statements is true?

- a) Physical changes form new substances. b) Chemical changes can always be reversed
c) Physical changes do not form new substances. d) Both a and b

Q6. When vinegar reacts with baking soda, it produces bubbles of gas. This shows that:

- a) It is a physical change. b) It is a chemical change. c) It is a reversible change. d) It is a temporary change

Q7. Which of the following changes is reversible?

- a) Burning of wood. b) Melting of wax. c) Cooking an egg. d) Rusting of iron

Q8. The melting of ice into water shows:

- a) New substance is formed. b) A chemical change. c) No new substance is formed. d) Both a and b

Q9. Which of the following shows both physical and chemical changes?

- a) Candle burning. b) Ice melting. c) Paper tearing. d) Water freezing

Q10. Which one of the following is not a sign of a chemical change?

- a) Change in color. b) Formation of gas. c) Change in temperature. d) Change in shape only